



Department of Computer Science and Business Systems

Academic Year 2024– 2025 (Odd/ ~~Even~~ Semester)

Degree, Semester & Branch: B.Tech, V & CSBS

Course Code & Title: CCS335 & Cloud Computing

Name of the Faculty member(s): Dr.M.Gomathy Nayagam

Innovative Practice Description

- **Unit/Topic:** Unit III / Introduction to Docker.

- **Course Outcome:** CO 3

- **Topic Learning Outcome:** 3b & 3c

- **Activity Chosen:** Flipped Class

- **Justification:**

Using flipped classroom active learning for the topic "Introduction to Docker" encourages students to learn foundational concepts independently through pre-class resources, such as videos and reading materials on Docker's fundamentals. In-class time is then dedicated to hands-on activities, such as creating Docker containers and experimenting with Docker commands. This approach maximizes active learning by allowing students to apply concepts in real-time, collaborate with peers, and receive immediate feedback from the instructor. The flipped model promotes deeper understanding, better retention, and more engagement, as students focus on practical, problem-solving tasks rather than passive content consumption.

- **Time Allotted for the Activity:**25 Minutes

- **Details of the Implementation:**

- Video lecture corresponding to the particular topic was hosted in online one week before.
- Students were instructed to watch the video and took the notes according to their understanding level in their home and come with preparation for discussion on one particular day in the class room.
- The students were segmented as a group in the class during the day of discussion.
- We gave 5 minutes for discussion among the group member regarding the video lecture they watched earlier.
- Each group collected all the important points related to the topic from those group members and to be allotted 15 minutes time for on board presentation and discussion about the topic in front of the whole class with their corresponding group members.
- Corresponding subject faculty monitored each group's presentation.

- Course Instructor selected one among the student in the class and asked the particular student to summarize all groups' presentation for next 5 minutes

- **CO-PO/ PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO 3	3	3	3	2	3	-	-	-	2	1	1	1	3	1	3
(1– Low 2–Moderate 3 – High)															

- **PO/ PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4		PO5
	3	3	3	2		2
Justification for correlation	The students could be able to apply the basic engineering knowledge know the Dockers concept	The students could be able formulate the solution for given problem using the Docker	The students could able to provide the solution for identifying the relationship between the Docker vs VM	The students could be able to apply Docker.		The students could be able to use the appropriate software's to create Docker
Innovative practice	PO9	PO10	PO11	PO12	PSO1	PSO2
	1	1	1	1	3	1
Justification for correlation	The students could able to know their responsibility to apply different Dockers techniques as an individual.	The students could be able to communicate their results of the problem and discuss with their peer team members	The students could able to apply the various configuration features of Dockers when they do mini project in this course	The students could be able to learn the newer Docker	The students could be able to apply Docker technology	The students could able to apply Dockertechnique in their application

Innovative practice	PSO3
	3
Justification for correlation	The student could able to apply the concept of Docker in real-time project

• **Images/Screen shot of the practice:**



Fig. Students Listen the NPTEL Video of Docker Concept



Fig. One of the Boys group discussed the points among themselves from the concepts they learned from the video.

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Students are enjoyed to prepare the concept through self-learning. They felt that this activity is very helpful to them for improving their self-learning ability.

❖ **Benefit of the Practice (E.g. Outcome Attainment would have increased due to innovative practice over conventional practice)**

- The students can able to explore the knowledge in map reduce programming model concepts by improving their self-paced learning and get more key points from their peer groups when they are doing collaborative discussion among them.

❖ ***Challenges faced in implementation:***

- Difficult to make the individual to watch the video which is already hosted in canvas.

❖ **References:**

- Dr.M.Gomathy Nayagam, “Map reduce” in Course CS8791 Cloud Computing, https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MGM_VII_Flipped-Class-Room.pdf

Signature of Faculty Member
(Dr.M.Gomathy Nayagam)

HOD